Formula 1
1950–2024

Data exploration and dashboard report

Final Project, Unit 3: Individual Data Exploration & Dashboard Development

Tuwaiq Academy, AI & Data Science Bootcamp

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1. Introduction

1.1 Dataset

Title	Formula 1 World Championship (1950–2024)
Source	Ergast-format F1 database, maintained by RaceOptiData: https://github.com/raceoptidata/Ergast_like-dumps (same tables as the Kaggle dataset https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020)
Format	Six related CSV files: results, races, drivers, constructors, circuits, status
Size	26,759 race results, 1,125 races, 864 drivers, 212 teams, 77 circuits

The dataset records every Formula 1 World Championship race from 1950 to the end of the 2024 season. The main file, *results*, has one row for every driver in every race, with the starting grid position, finishing position, points, laps completed and a status code that says how the race ended. The other files describe the races, drivers, teams, circuits and statuses that the codes refer to.

Why I chose it. Formula 1 combines sport, engineering and 75 years of history, so the data can answer questions about both people and machines. It also comes as several related tables rather than one file, so building the analysis table required merging them, which is a skill most single-file datasets do not practise.

1.2 Problem definition

The project explores how Formula 1 has changed since 1950 and what decides who wins. The main questions are:

- How has the sport grown (races per season, drivers, circuits)?
- Have cars become more reliable, and why do cars fail to finish?
- Which drivers and teams have won the most, and how dominant can one team be?
- How important is the starting position, and has it become more important?
- Are race winners getting younger?
- Which countries produce the most drivers?

Expected insights: more races per season today; far fewer retirements; Hamilton, Schumacher and Ferrari at the top of the win tables; and a clear advantage for cars that start at the front.

2. Data exploration

All six files were loaded with pandas directly from GitHub. Missing values are stored as the text *NULL* in these files, so they were read as proper missing values with *na_values="NULL"*.

File	Rows	Columns	Contents
results	26,759	18	One row per driver per race
races	1,125	18	Season, round, circuit, date
drivers	864	9	Name, birth date, nationality
constructors	212	5	Team name and nationality
circuits	77	9	Name, country, latitude, longitude
status	141	2	How a race ended (Finished, Engine, ...)

2.1 Structure of the results table

df.info() shows 18 columns: 15 numeric and 3 text. The ID columns, grid, points, laps and status are complete. *df.describe()* for the main numeric columns:

	grid	position	positionOrder	points	laps
count	26,759	15,806	26,759	26,759	26,759
mean	11.1	8.0	12.8	2.0	46.3
std	7.2	4.8	7.7	4.4	29.5
min	0	1	1	0	0
median	11	8	12	0	53
max	34	33	39	50	200

2.2 Missing values and quality checks

Column	Missing	%	Meaning
position	10,953	40.9	Empty when the driver did not finish or was not classified
time, milliseconds	19,079	71.3	Only recorded for the leading finishers
fastestLap, rank, ...	~18,500	~69	Lap timing only exists from 2004
number	6	0.0	Car number missing in a few early entries

Other checks: there are **no duplicate rows**. About **1,600 entries never started a race** (did not qualify, did not pre-qualify, withdrew). **grid = 0** means a pit-lane start, not first place. There are **141 different statuses**, too many to compare directly, and the most common is Finished, followed by +1 Lap, Engine, +2 Laps and Accident.

3. Data cleaning

Step	What was done	Why
Merge	Joined the six tables on raceId, driverId, constructorId, circuitId and statusId	One table with readable names instead of ID codes
Rename	positionOrder to position; forename + surname to driver; name columns to race, team, circuit	Clear, consistent column names
Remove rows	Dropped 1,604 entries that never started (did not qualify / pre-qualify, withdrew)	They are not race results
Fix values	grid = 0 replaced with missing (130 rows after the step above)	0 means pit-lane start, not pole
Convert types	date and dob converted to datetime	Needed to calculate driver age
Duplicates	drop_duplicates(): none found	Data was already unique
Group statuses	141 statuses to Finished, Mechanical, Accident, Other	Makes results comparable
Save	Cleaned table saved to clean_data.csv (25,155 rows × 23 columns)	Used by the dashboard

New columns were added for the analysis: *decade*, *age* (driver age on race day), and the flags *finished*, *win*, *podium* (top 3) and *pole* (started first). After grouping, the result types are: Finished 15,161 entries, Mechanical 6,714, Accident 2,777 and Other 503.

4. Exploratory data analysis

This section shows the main graphs, from single-variable distributions to relationships between variables. For the six main questions from the problem definition, a hypothesis was written before making the graph and checked afterwards.

4.1 Univariate analysis

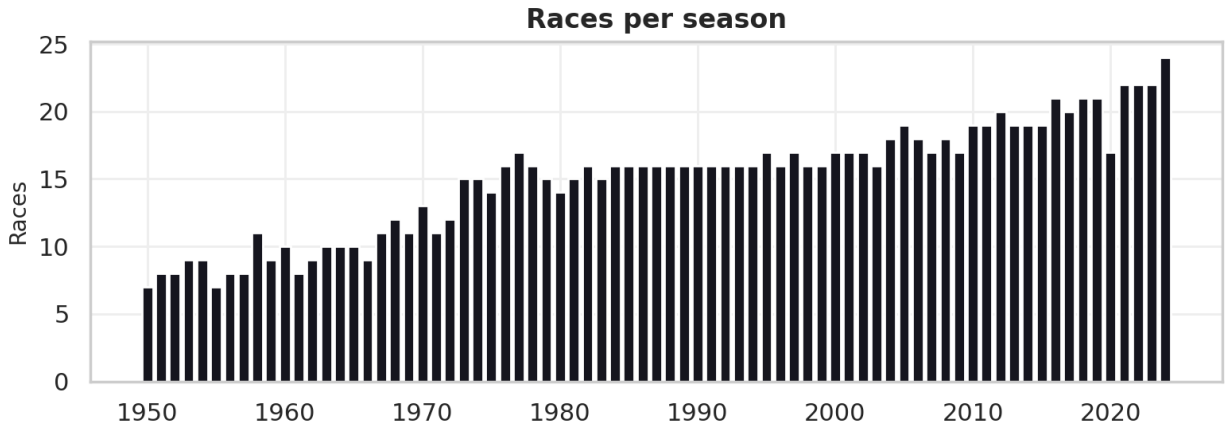


Figure 1. Races per season. The calendar grew from 7 races in 1950 to 24 in 2024, more than three times bigger, with the fastest growth after 2000.

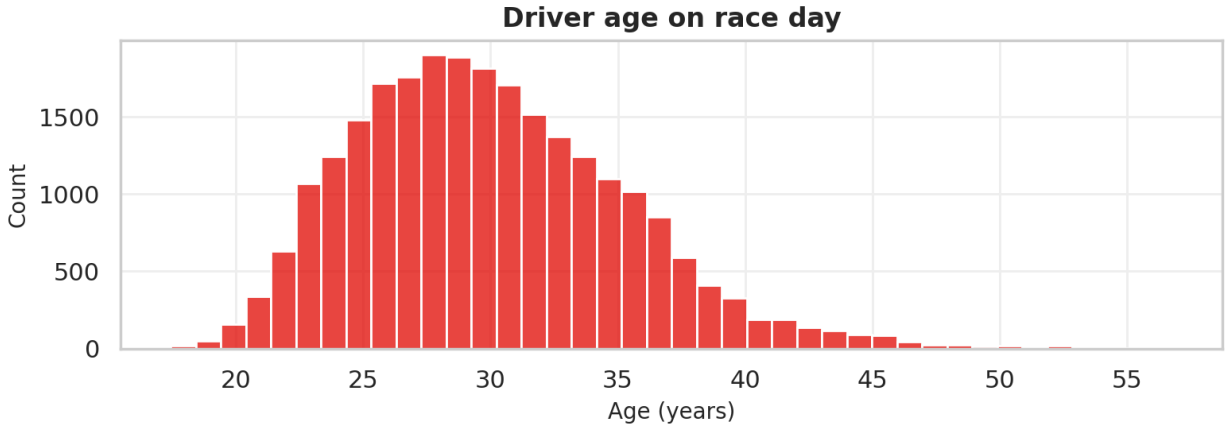


Figure 2. Driver age on race day. Most entries are by drivers aged 26 to 33 (median 29.4). Ages above 44 are outliers (1.3% of entries), and 99% of them are from before 1975, when older drivers were common.

Hypothesis: most cars finish the race, and when they do not, accidents are the main reason.

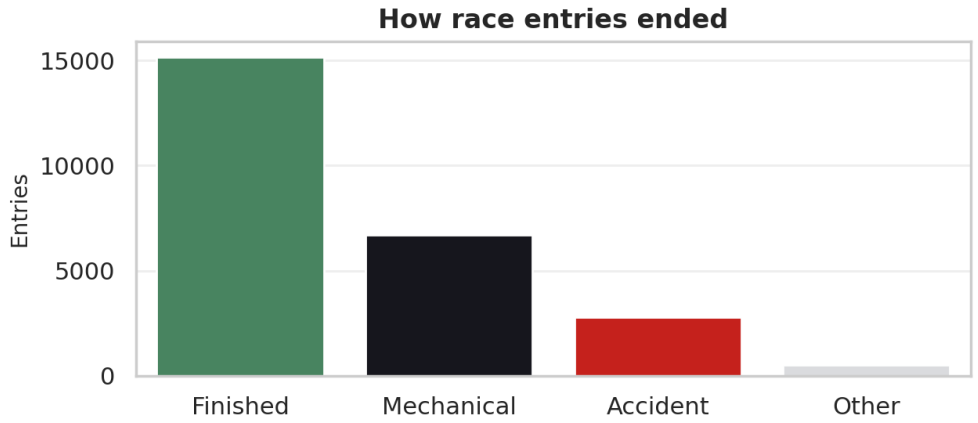


Figure 3. How race entries ended. **Partly true:** 60% of entries finished, but the main reason for not finishing is mechanical failure (27%), not accidents (11%).

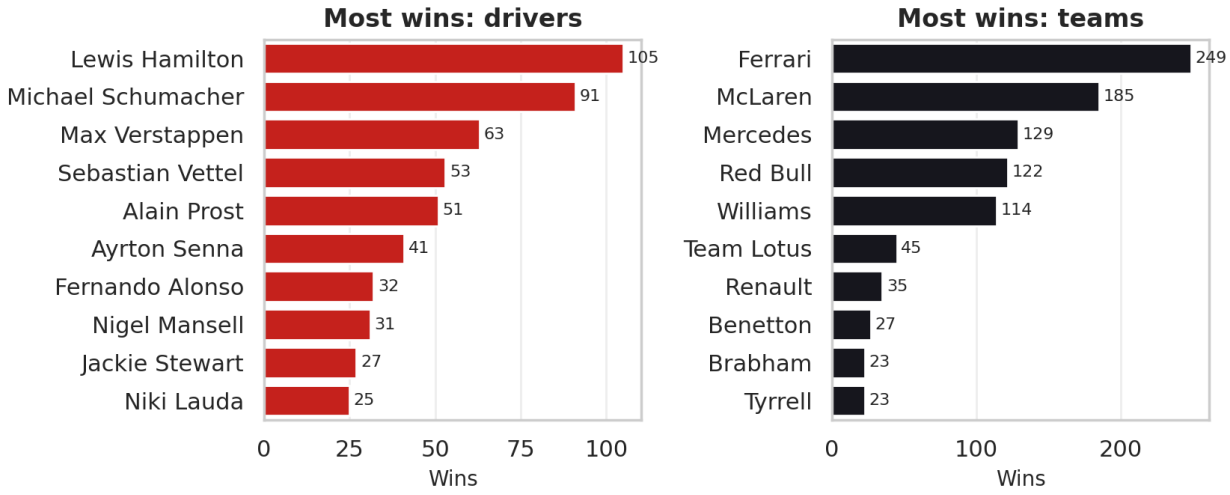


Figure 4. Most race wins. Hamilton leads with 105 wins ahead of Schumacher (91) and Verstappen (63). Ferrari has 249 wins, well ahead of McLaren (185) and Mercedes (129). Only 115 of 789 drivers have ever won a race.

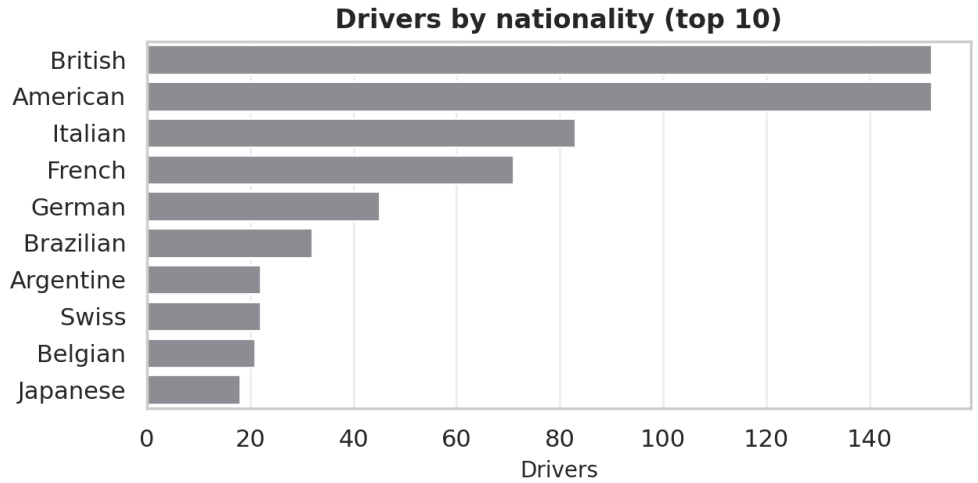


Figure 5. Drivers by nationality. Britain and the USA are tied with 152 drivers each. The US number is high because the Indianapolis 500 counted towards the championship from 1950 to 1960.

4.2 Bivariate and multivariate analysis

Hypothesis: cars have become more reliable, so the share of cars that finish should rise over time.

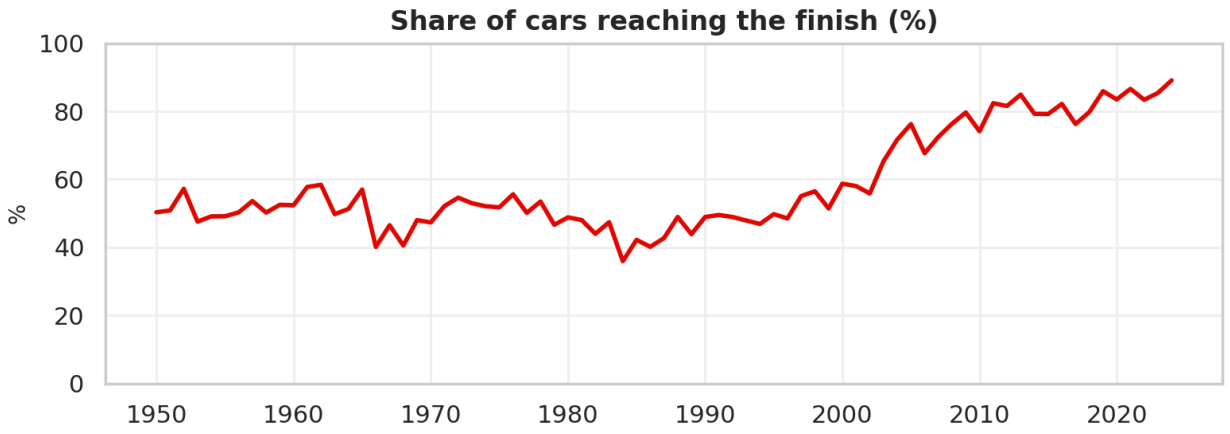


Figure 6. Share of cars reaching the finish. **Partly true:** the finish rate stayed around 50% until the late 1990s and even fell to its lowest point in 1984 (36%). The improvement only started around 2000, reaching 89% in 2024.

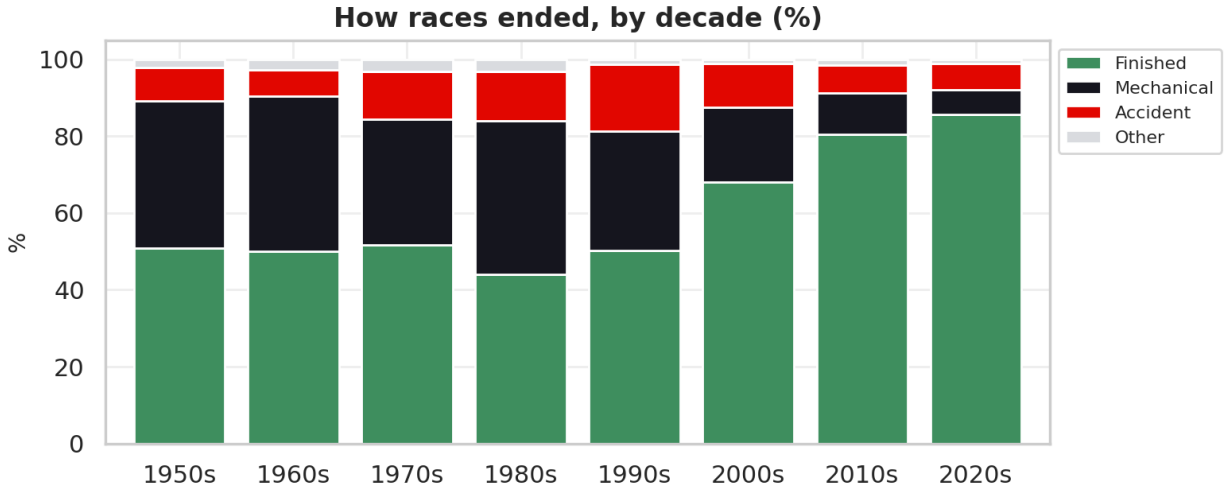


Figure 7. How races ended, by decade. This explains Figure 6: mechanical failures were 30–40% of entries from the 1950s to the 1990s and only 6% in the 2020s. Accidents peaked in the 1990s (17%).

Hypothesis: starting from pole gives the best chance of winning, and the chance drops quickly further back.

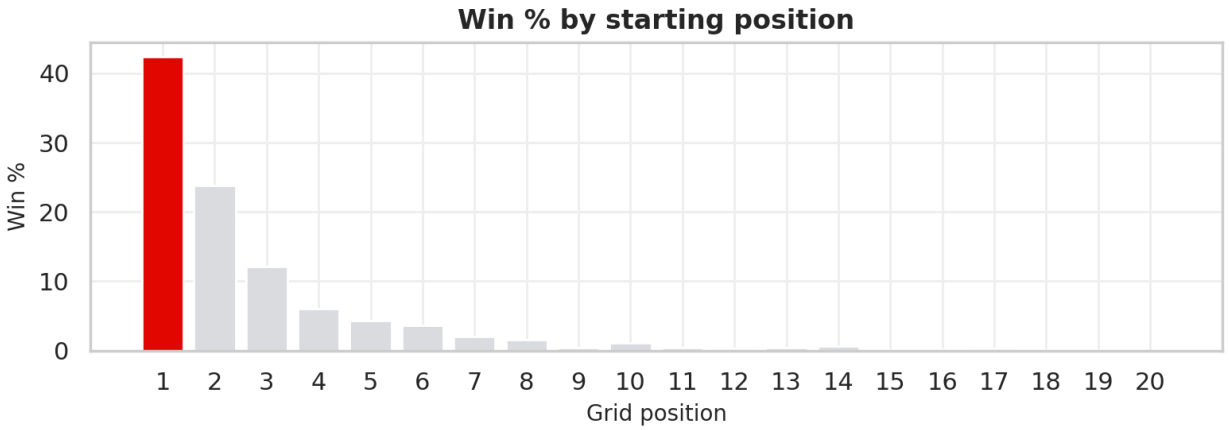


Figure 8. Win % by starting position. **Confirmed:** pole wins 42% of races, 2nd place 24% and 3rd 12%. From 5th place back the chance is under 5%.

Hypothesis: overtaking is harder in modern F1, so more races are won from pole than in the past.

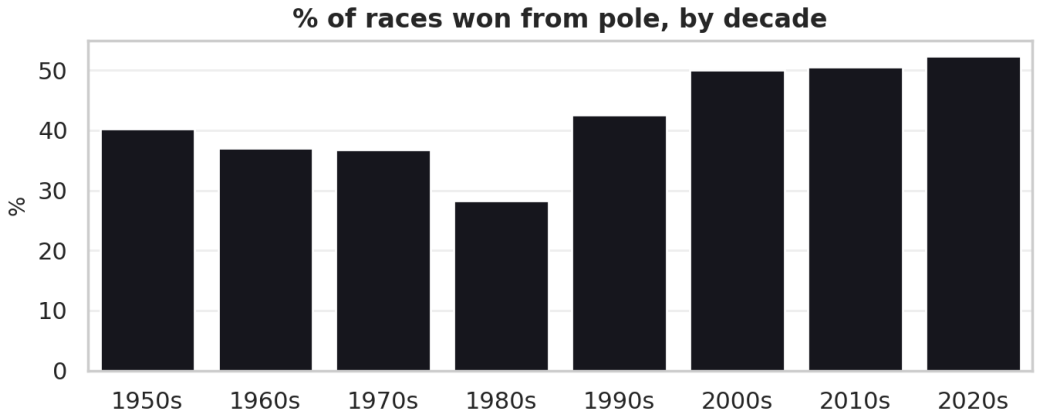


Figure 9. Share of races won from pole, by decade. **Confirmed:** only 28% in the 1980s, but about 50% in every decade since 2000. Qualifying has become much more important.

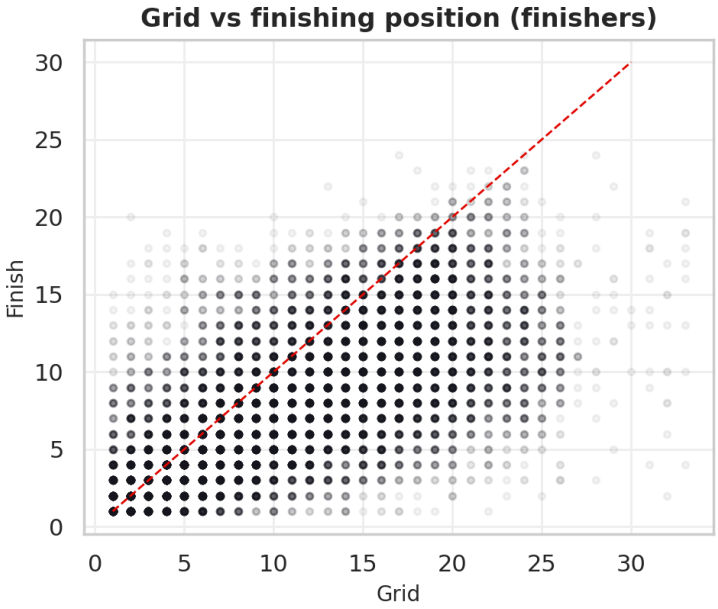


Figure 10. Grid position against finishing position for cars that finished. Most points sit near the diagonal (correlation 0.68): cars usually finish close to where they start.

Hypothesis: in some seasons, one team won almost every race.

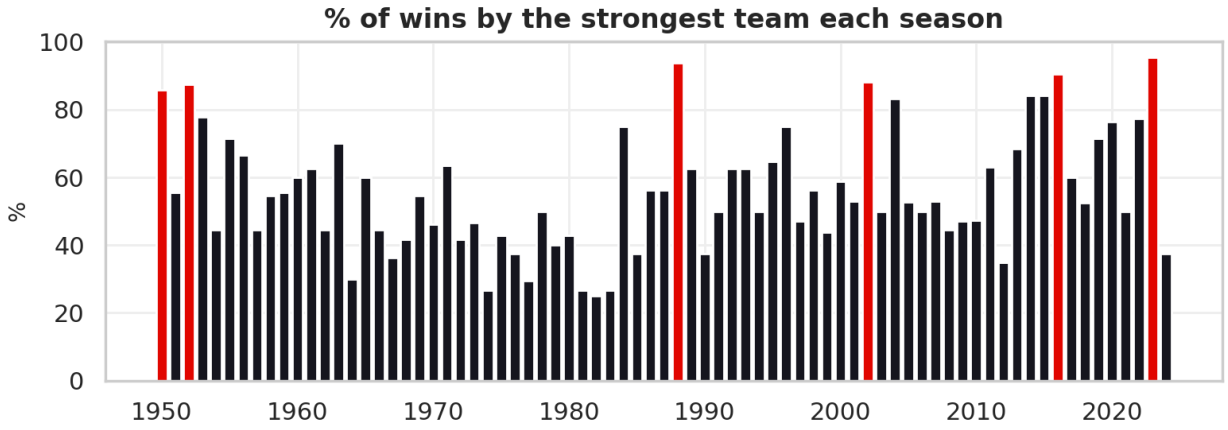


Figure 11. Share of the season's wins taken by the strongest team (seasons above 85% in red). **Confirmed:** Red Bull won 95% of races in 2023 (21 of 22), McLaren 94% in 1988 and Mercedes 90% in 2016.

Hypothesis: race winners are younger today than in the past.

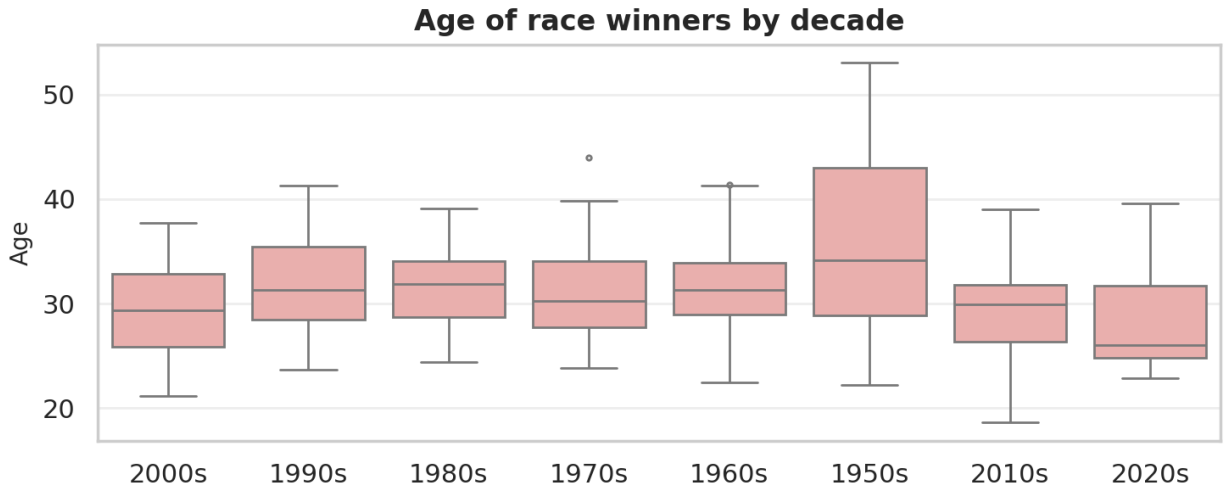


Figure 12. Age of race winners by decade. **Confirmed:** the median winner was 34 in the 1950s and 26 in the 2020s, the youngest decade by far.

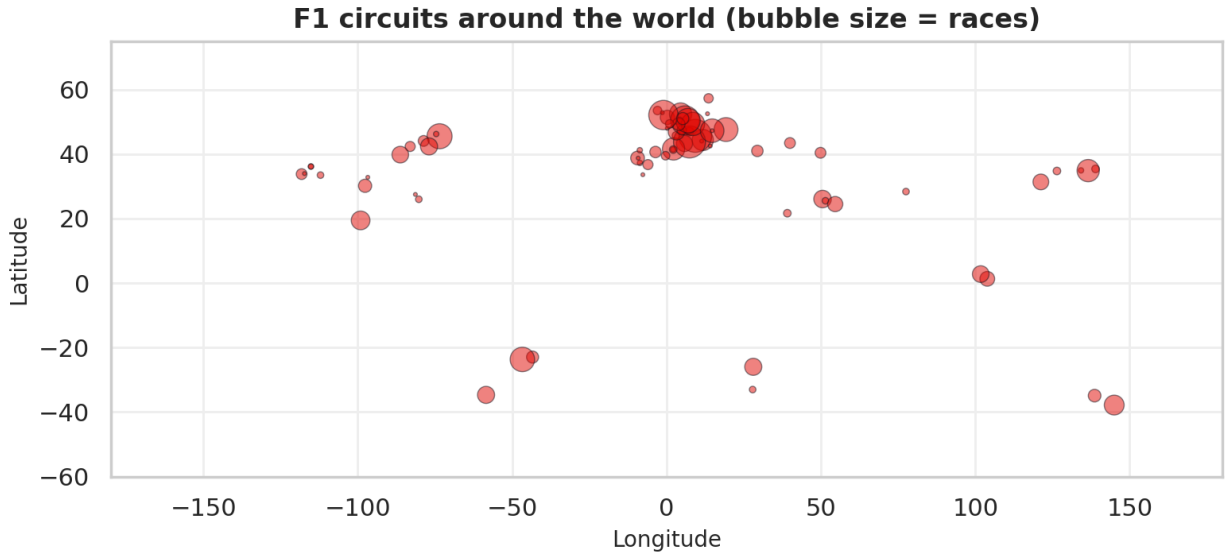


Figure 13. F1 circuits around the world, sized by number of races. About 60% of all races were held in Europe; Monza (74 races) and Monaco (70) have hosted the most.

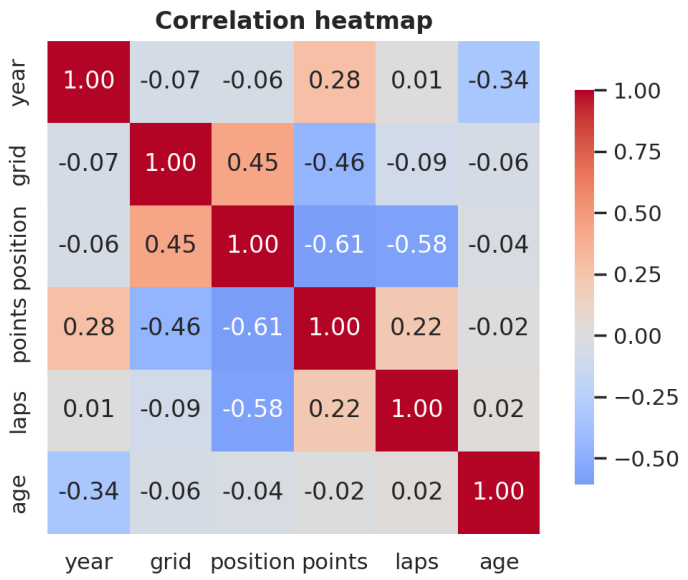


Figure 14. Correlation heatmap. Finishing position is most strongly linked to points (-0.61) and laps (-0.58). Grid is linked to position (0.45) and points (-0.46). Year and age are negatively correlated (-0.34), so drivers got younger over time.

4.3 Predictive model

A random forest was trained to predict whether a driver finishes on the podium from the starting position, the season and the driver's age, using an 80/20 train-test split.

Model	Accuracy	Podium precision	Podium recall
Always guess 'no podium'	0.872	-	0.00
Random forest (grid, year, age)	0.891	0.64	0.36

The model is only slightly better than the baseline and finds about a third of podiums. It shows that the grid position is a strong signal (a 2024 driver starting 1st gets a 77% podium chance, starting 10th only 6%), but without information about the team and car it cannot predict podiums reliably.

5. Dashboard overview

The interactive dashboard was built with Streamlit and Plotly (*app.py*). The sidebar contains the dataset description and three filters: a **slider** for the season range, a **dropdown** for the team and a **multiselect** for driver nationality. Every number, chart and insight line on the page updates with the filters. The main page has four headline numbers and eight tabs.

Tab	Contents
Overview	Races per season, finish rate over time, how races ended by decade
Drivers & teams	Most wins by driver and team, winner age by decade, drivers by nationality
Starting grid	Win % by grid position, pole wins by decade, team dominance per season
Circuits	World map of circuits and the most used tracks
Driver lookup	Pick any driver: starts, wins, podiums, poles, seasons and teams
Podium predictor	The trained model: choose a grid slot, season and age to get a podium chance
Data	Data preview, summary statistics and a CSV download of the current selection
Key findings	Main insights and limitations

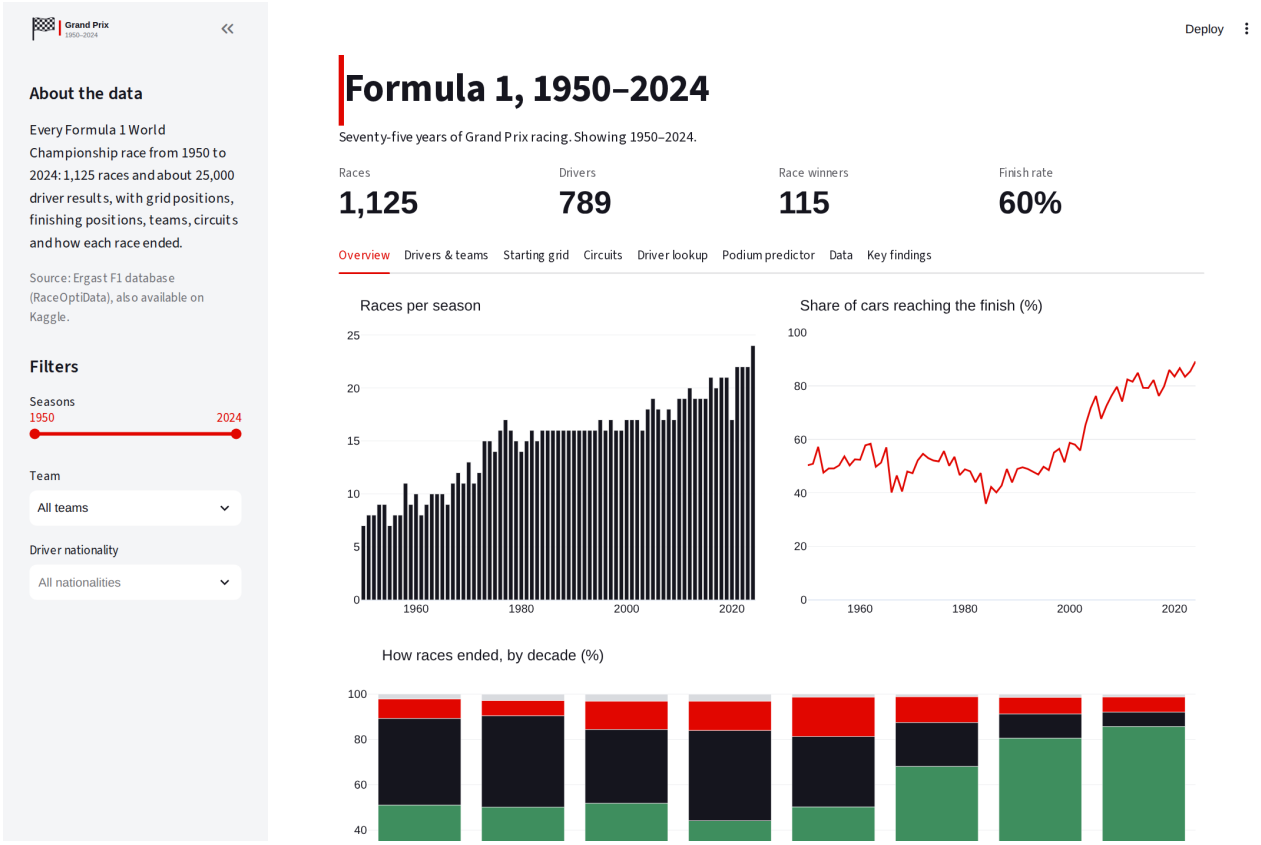


Figure 15. Dashboard, Overview tab with the sidebar filters and headline numbers.

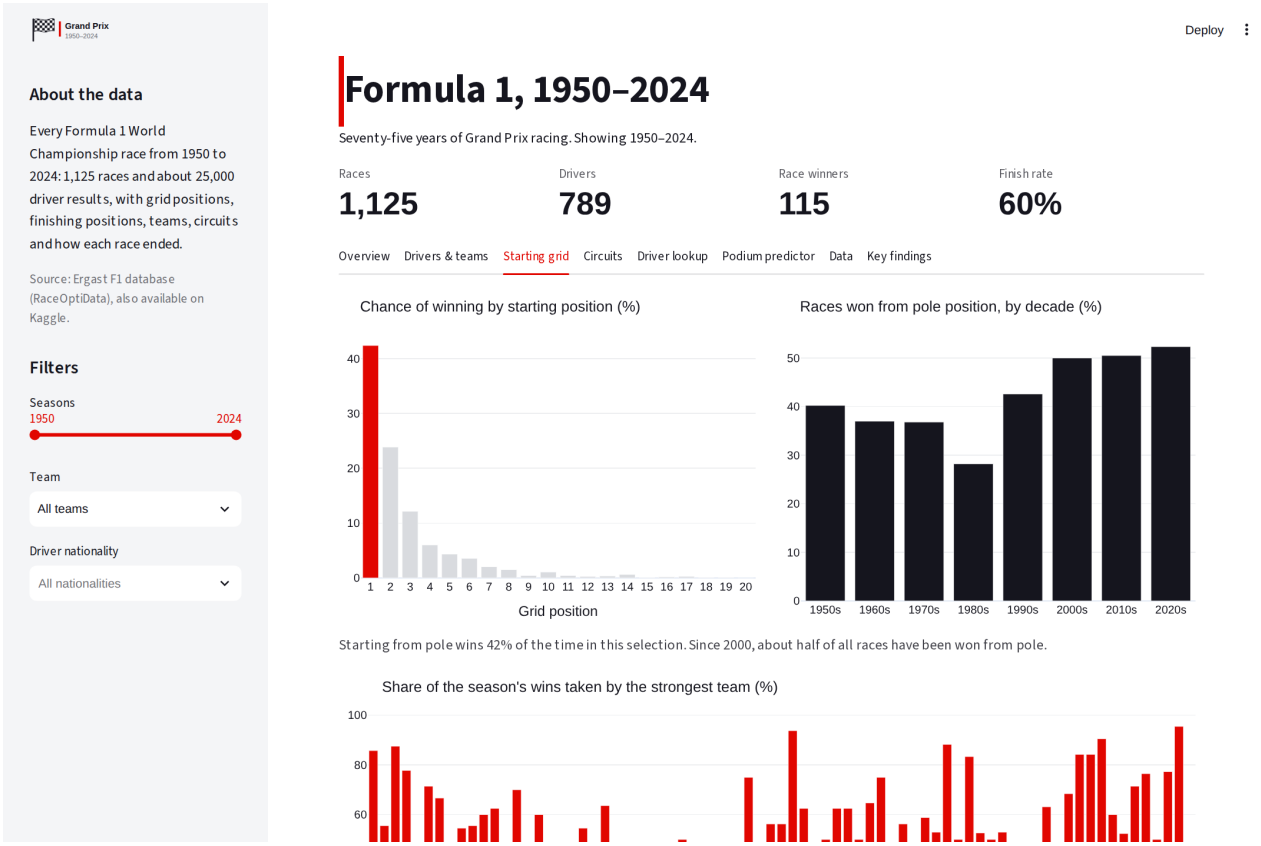


Figure 16. Dashboard, Starting grid tab. The pole position bar is highlighted.

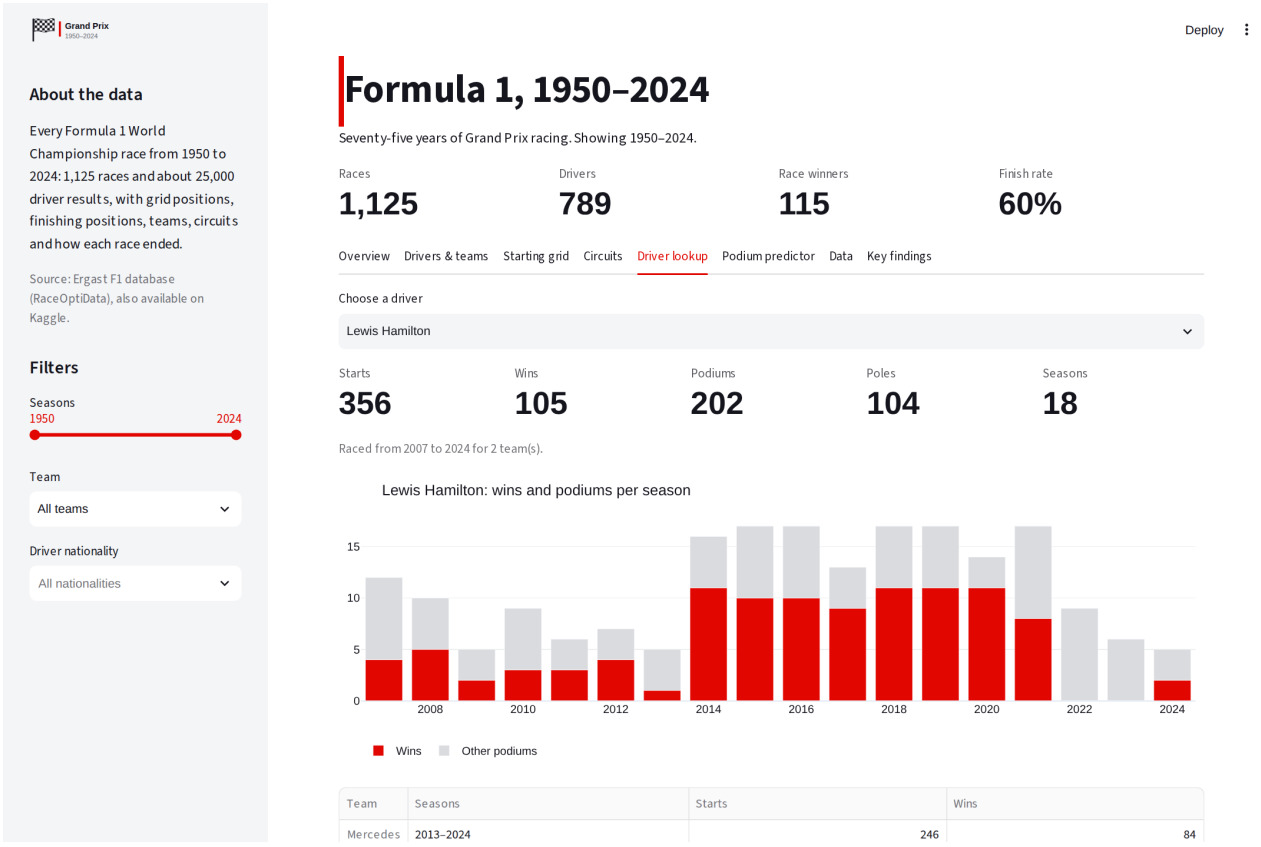


Figure 17. Dashboard, Driver lookup tab (Lewis Hamilton).

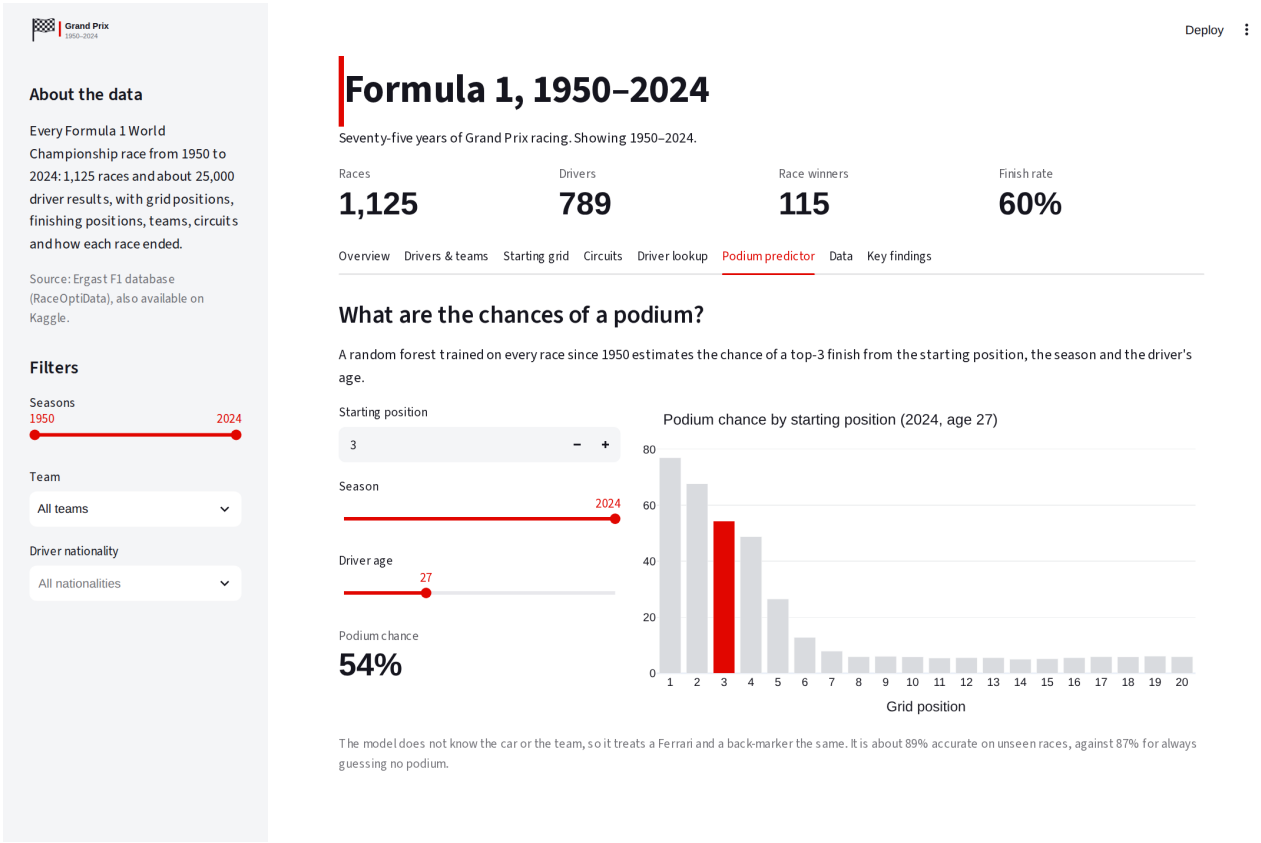


Figure 18. Dashboard, Podium predictor tab. The selected grid slot is highlighted in the chart.

6. Insights and conclusion

6.1 Main insights

- **Reliability transformed the sport.** For almost 50 years only about half the cars finished, mostly because of mechanical failures. In 2024, 89% of starters reached the flag.
- **Qualifying became decisive.** Pole wins 42% of races overall, and about half of all races since 2000.
- **Success is concentrated.** Only 115 of 789 drivers have won a race, and single teams can win almost a whole season (Red Bull 2023: 95%).
- **Winners are younger.** The median winner went from 34 in the 1950s to 26 in the 2020s.
- **F1 grew and spread.** From 7 to 24 races per season, with Europe still hosting about 60% of all races.

6.2 What I learned

- How to merge several related tables into one analysis table using ID columns.
- Missing values are not always errors: an empty finishing position simply means the driver did not finish.
- Grouping many categories (141 statuses) into a few makes them far easier to analyse.
- A model must be compared with a baseline: a simple rule can be almost as accurate.

6.3 Limitations of the dataset

- Rules, points systems, race lengths and field sizes changed many times, so eras are not directly comparable. Wins and podiums were used instead of points for that reason.
- The Indianapolis 500 (1950–1960) was a different kind of race but is counted in the championship data.
- There is no weather or tyre data, and lap times only exist from 2004.
- Teams that changed names or owners (for example Brawn to Mercedes) are counted as different teams.

6.4 Recommendations for future analysis

- Use the lap-time and pit-stop tables (2004 onwards) to study race strategy.
- Add weather data to see how rain changes results.
- Add team strength to the model, for example the team's results in the previous season.
- Compare teammates in qualifying to rank drivers more fairly across eras.

7. References

- RaceOptiData. Ergast-like F1 data dumps (CC BY-SA 4.0). https://github.com/raceoptidata/Ergast_like-dumps
- Rao, R. Formula 1 World Championship (1950–2020). Kaggle. https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020